Project Proposal

Context-Aware Regulatory Change Impact Analysis for Small Food Manufacturers in Sri Lanka

Proposed System Name

RegImpact-SL

Proposed Research Area

Artificial Intelligence, Information Retrieval, Retrieval-Augmented Generation, Regulatory Intelligence, Business Information Systems, Natural Language Processing, and AI-Assisted Decision Support

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1. Introduction

Small and medium-sized enterprises (SMEs) play a major role in Sri Lanka's economy. Recent Sri Lankan and Asian Development Bank sources describe SMEs as a major contributor to employment and GDP, while also identifying regulatory, institutional, and business-environment constraints affecting their growth.

For a small food manufacturing business, regulatory compliance is not represented by a single regulation or government institution. A business may need to consider food regulations, manufacturing registration, environmental requirements, labour-related obligations, taxation, local-authority requirements, and product-specific requirements.

The difficulty becomes greater when regulations change.

A business owner may need to determine:

1. What regulation has changed?
2. When does the change become effective?
3. Does the change apply to my business?
4. Why does it apply or not apply?
5. Which business activity, product, process, or document is affected?
6. What action should the business take?
7. What authoritative evidence supports the conclusion?

Existing AI systems can retrieve regulatory documents or generate answers using large language models. However, simply retrieving documents is insufficient for business-specific regulatory impact analysis because applicability can depend on contextual variables such as employee count, product type, production capacity, business activity, import/export status, and effective dates.

This project therefore proposes **RegImpact-SL**, a context-aware AI system designed to identify and explain the business-specific impact of regulatory changes on small food manufacturers in Sri Lanka.

The central idea is:

Business Context + Regulatory Change → Applicability → Impact → Action → Evidence

The proposed system is not intended to provide legal advice or replace regulatory authorities. It is an AI-assisted decision-support and research prototype designed to demonstrate how structured business context, regulatory knowledge, hybrid information retrieval, and evidence-grounded language models can support regulatory impact analysis.

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2. Problem Statement

Sri Lankan small food manufacturers operate within a fragmented regulatory environment involving multiple authorities and regulatory instruments.

For example, official sources currently contain food regulations covering areas such as labelling, trans-fat, colour coding, iodization, registration of food premises, and other food-control requirements. The Food Control Administration Unit maintains a collection of current food regulations and describes a domestic control process involving application, documentation, inspection, sampling, laboratory testing, and certification.

Manufacturing businesses may also have separate industry-registration obligations. The Ministry of Industry states that manufacturing industry production sites generally need to be registered within three months of commencement, subject to stated exceptions. Industry registration can also involve documents such as business registration, environmental protection licensing, EPF/ETF documentation, machinery information, process-flow information, and financial information.

Environmental requirements can introduce additional conditionality. For example, the Central Environmental Authority's Environmental Protection Licence framework uses prescribed-activity classifications and includes activities for which licensing responsibility can involve local authorities.

Consequently, the problem is not simply:

"Find information about food regulations."

The more difficult problem is:

Determine how a regulatory change affects a specific business given its characteristics and operating context.

For example, two businesses may operate in the same industry but have different employee counts, products, production capacities, or import/export activities. Therefore, the same regulatory provision may have different applicability or operational consequences.

A conventional keyword-search system may retrieve the regulation but fail to determine its business-specific relevance.

A conventional vector RAG system may retrieve semantically relevant passages but still fail when applicability depends on exact thresholds, conditions, dates, or combinations of business attributes.

This creates an opportunity to investigate whether **business-context-aware regulatory retrieval and structured reasoning can improve regulatory impact analysis compared with conventional retrieval approaches.**

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3. Research Question

Main Research Question

How can a context-aware AI system identify and explain the business-specific impact of regulatory changes for small food manufacturers in Sri Lanka?

Supporting Research Questions

RQ1

How accurately can a context-aware system determine whether a regulatory change applies to a specific small food manufacturer?

RQ2

How accurately can the system identify the obligations and operational impacts resulting from an applicable regulatory change?

RQ3

Does structured business context combined with hybrid retrieval improve evidence-grounded regulatory impact analysis compared with conventional RAG?

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4. Research Hypothesis

Alternative Hypothesis — H1

Incorporating structured business context into regulatory retrieval and reasoning improves the accuracy of regulatory applicability and impact identification compared with conventional RAG.

Null Hypothesis — H0

Incorporating structured business context does not produce a statistically or practically meaningful improvement in regulatory applicability and impact identification compared with conventional RAG.

The hypothesis will be evaluated experimentally using a controlled benchmark containing multiple business-regulation scenarios.

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5. Aim

The primary aim of this project is to design, implement, and experimentally evaluate a context-aware AI system capable of analysing regulatory changes and identifying their business-specific applicability, impact, required actions, and supporting evidence for small food manufacturers in Sri Lanka.

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6. Objectives

Objective 1 — Build a Regulatory Knowledge Base

Collect and structure authoritative regulatory information relevant to small food manufacturers.

The knowledge base will capture:

- regulation identity
- regulatory authority
- version
- publication date
- effective date
- applicability conditions
- thresholds
- obligations
- affected activities
- regulatory changes
- source documents
- section/clause information
- page-level evidence

Objective 2 — Represent Business Context

Develop a structured representation of a small food manufacturer's characteristics, including:

- industry
- number of employees
- products
- production capacity
- business activities
- premises
- domestic/import/export activity
- relevant operational characteristics
- other attributes required for regulatory applicability.

Objective 3 — Detect and Represent Regulatory Changes

Identify meaningful changes between regulatory versions, such as:

- changed thresholds
- new obligations
- removed obligations
- changed applicability conditions
- changed effective dates
- amended requirements.

Objective 4 — Develop Multiple Retrieval Baselines

Implement and evaluate:

1. keyword retrieval
2. vector-based retrieval/RAG
3. hybrid retrieval/RAG
4. proposed context-aware retrieval and reasoning approach.

This allows the research contribution to be evaluated experimentally rather than simply demonstrated through a working application.

Objective 5 — Develop Context-Aware Applicability Reasoning

Determine whether a regulatory change applies to a particular business by comparing:

Business Context

against

Regulatory Conditions

The system should be capable of distinguishing scenarios such as:

- 14 employees vs. 15 employees
- domestic vs. export business
- one product category vs. another
- production below vs. above a threshold
- applicable vs. non-applicable business activities.

Objective 6 — Identify Business Impact

For applicable changes, determine the affected:

- operations
- products
- documentation
- packaging
- labelling
- licensing
- reporting
- production processes
- employee-related processes
- compliance activities.

Objective 7 — Generate Evidence-Grounded Actions

The system should generate understandable recommendations such as:

- review a label

- update documentation
- register an activity
- review an operational process
- monitor an effective date
- investigate an additional requirement.

Every important conclusion should be linked to authoritative evidence.

Objective 8 — Evaluate the Proposed Approach

Evaluate the proposed system against baseline approaches using quantitative and, where feasible, human-centred evaluation.

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7. Scope

The initial scope is deliberately restricted to:

Geographic scope

Sri Lanka

Business scope

Small food manufacturing businesses

Regulatory scope

Selected regulations relevant to:

- food control
- food labelling
- manufacturing
- environmental requirements
- labour-related obligations
- taxation/business requirements
- selected local-authority requirements.

Core task

Regulatory change impact analysis.

Core output

The system will produce:

CHANGE → APPLICABILITY → REASON → IMPACT → ACTION → EFFECTIVE DATE → EVIDENCE

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8. Example Use Case

Consider a hypothetical small food manufacturer:

Business ID: B01

- Employees: 6
- Industry: food manufacturing
- Products: packaged biscuits/confectionery
- Production: below 250 kg flour/day
- Market: domestic
- Importing: No
- Exporting: No

A regulatory change is introduced.

The system processes:

Business Profile

+

Regulatory Change

It then determines:

1. What changed?

2. Does it apply?

3. Why?

4. What obligation is affected?

5. What business activity is affected?

6. What action should be taken?

7. When does it become effective?

8. What evidence supports the conclusion?

The final result could conceptually look like:

REGULATORY CHANGE

New/changed requirement

APPLICABILITY

Applicable

REASON

The business satisfies the relevant

activity/context conditions.

BUSINESS IMPACT

Product packaging / documentation

REQUIRED ACTION

Review affected materials/processes.

EFFECTIVE DATE

[Verified regulatory date]

EVIDENCE

Official regulation

Section / clause

Page

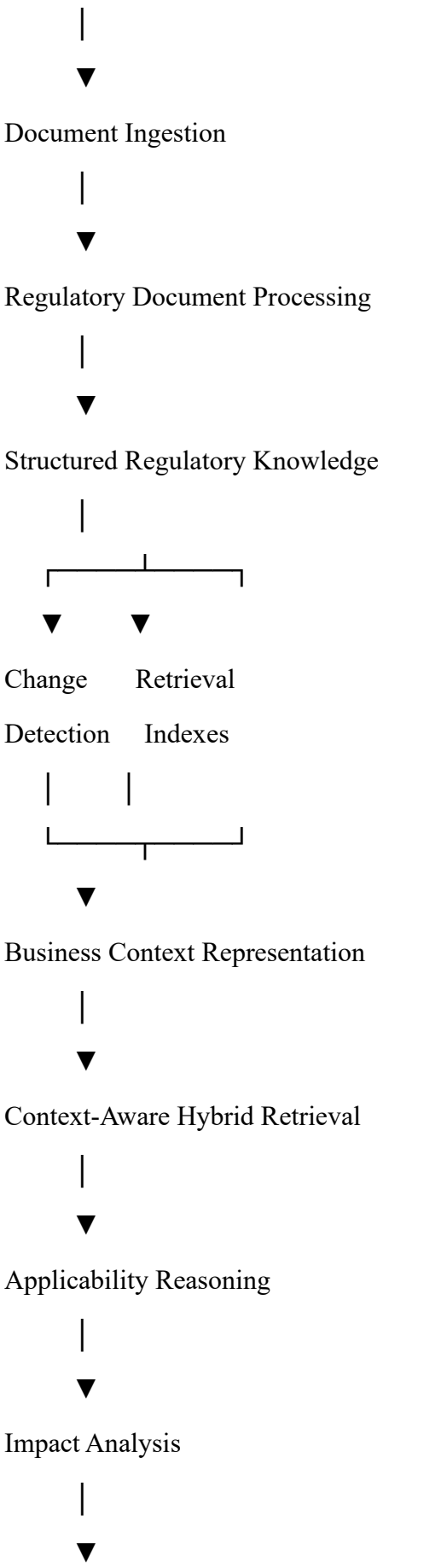
The system will explicitly distinguish between **verified evidence** and uncertainty rather than presenting unsupported AI-generated claims as facts.

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9. Proposed System Architecture

The proposed architecture consists of the following major layers.

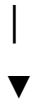
Official Regulatory Sources


Document Ingestion
Regulatory Document Processing
Structured Regulatory Knowledge
Change Detection
Retrieval Indexes
Business Context Representation
Context-Aware Hybrid Retrieval
Applicability Reasoning
Impact Analysis

Action Generation



Evidence Verification



Structured Impact Report

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10. Regulatory Document Processing

Authoritative documents will be processed through a controlled pipeline:

Official PDF

↓

Text Extraction

↓

Cleaning

↓

Section/Clause Detection

↓

Metadata Extraction

↓

Document Versioning

↓

Evidence Indexing

The system will preserve information such as:

- source document
- regulation identifier
- version
- section
- clause

- page
- publication date
- effective date
- source authority.

This is necessary because regulatory AI must be able to answer not only:

"What is the answer?"

but also:

"Where did the answer come from?"

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11. Regulatory Knowledge Representation

The system will use structured representations such as:

Regulation

├── Identity

├── Authority

├── Version

├── Publication Date

├── Effective Date

├── Applicability

├── Conditions

├── Thresholds

├── Obligations

├── Affected Activities

└── Evidence

The project will initially use structured data rather than immediately introducing a full knowledge graph.

A knowledge graph may later be evaluated as an experimental extension if the results justify it.

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12. Business Context Representation

Business information will be represented as structured variables.

Example:

Business

├── Industry

├── Employees

├── Products

├── Production Capacity

├── Location

├── Activities

├── Import Status

├── Export Status

└── Other Relevant Attributes

This enables the system to perform explicit contextual matching rather than relying entirely on natural-language similarity.

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13. Retrieval Architecture

The proposed system will use hybrid retrieval.

Keyword Retrieval

Useful for:

- exact legal terminology
- Gazette numbers
- section numbers
- thresholds
- dates
- named requirements.

Semantic Retrieval

Useful when the user's wording differs from the wording in the regulation.

Metadata Filtering

Useful for:

- regulation authority

- effective date
- business activity
- regulation type
- version.

Reranking

Candidate documents/passages will be reranked to identify the most relevant evidence.

Therefore:

Keyword Retrieval

+

Semantic Retrieval

+

Metadata Filtering

↓

Candidate Evidence

↓

Reranking

↓

Relevant Regulatory Evidence

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14. Applicability Reasoning

The applicability engine is a core research component.

It will compare structured business attributes against regulatory conditions.

For example:

Regulation condition:

Employees >= 15

Business:

Employees = 14

Result:

Not applicable

Changing only one variable:

Employees = 15

Result:

Applicable

Such minimal-pair scenarios are particularly important because they test whether the system genuinely uses business context rather than merely retrieving similar documents.

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15. Impact Analysis

After determining applicability, the system will identify the resulting business impact.

Conceptually:

Regulatory Change

↓

Applicable Obligation

↓

Affected Business Activity

↓

Operational Impact

Possible impact categories include:

- product
- packaging
- labelling
- manufacturing
- documentation
- licensing
- reporting
- employee processes

- quality-control processes.

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16. Evidence Verification

Evidence verification will be treated as a separate stage.

The system should associate claims with:

Claim

↓

Regulation

↓

Section/Clause

↓

Page

↓

Official Source

This is intended to reduce unsupported claims and hallucinations.

The project will measure evidence quality rather than assuming that an LLM-generated citation is automatically correct.

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17. Research Methodology

The project will follow an experimental design.

Phase 1 — Foundation

- project structure
- domain models
- dataset loaders
- validation
- benchmark framework.

Phase 2 — Knowledge Representation

- regulatory schemas
- business-context schemas

- obligation representation
- applicability conditions.

Phase 3 — Document Processing

- official-document ingestion
- text extraction
- cleaning
- section detection
- metadata preservation.

Phase 4 — Baseline 1

Keyword retrieval

This establishes a simple baseline.

Phase 5 — Baseline 2

Vector RAG

Documents will be embedded and semantically retrieved.

Phase 6 — Baseline 3

Hybrid RAG

Keyword and semantic retrieval will be combined.

Phase 7 — Proposed System

Business Context + Hybrid Retrieval + Structured Regulatory Reasoning + LLM

Phase 8 — Evaluation

Compare all approaches using the research benchmark.

Phase 9 — Engineering System

Develop the polished application:

- API
- frontend
- database
- authentication if required
- logging
- testing

- Docker
- CI/CD.

Phase 10 — Research Output

Produce:

- final report
- research paper
- methodology
- experiments
- results
- limitations
- future research.

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18. Dataset Strategy

The project will contain two major datasets.

Dataset A — Regulatory Corpus

Target initial size:

50–100 authoritative regulatory documents/versions

The corpus will prioritize official sources.

Potential sources include:

- Ministry of Health / FCAU
- Ministry of Industry
- Central Environmental Authority
- Department of Labour
- ETF
- IRD
- relevant local authorities
- official Gazette publications.

The FCAU's current regulatory catalogue already demonstrates that the food domain contains multiple regulations and versions, including 2026 labelling regulations and earlier instruments.

Dataset B — Regulatory Impact Benchmark

Target:

100–300 business-regulation scenarios

Each scenario will contain:

Business Profile

+

Regulatory Change

+

Ground Truth Applicability

+

Reason

+

Affected Obligation

+

Business Impact

+

Required Action

+

Evidence

The benchmark will contain both positive and negative examples.

It will also contain controlled minimal-pair scenarios.

Example:

Scenario A:

14 employees

Scenario B:

15 employees

This enables controlled evaluation of contextual reasoning.

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19. Baseline Experiments

The project will compare four major approaches.

Baseline 1 — Keyword Search

Query

↓

Keyword Search

↓

Relevant Documents

Baseline 2 — Vector RAG

Query

↓

Embedding

↓

Vector Search

↓

LLM

Baseline 3 — Hybrid RAG

Query

↓

Keyword + Vector Search

↓

Reranking

↓

LLM

Proposed Approach

Business Context

+

Regulatory Change

↓

Context-aware Retrieval

↓

Structured Applicability Reasoning

↓

Impact Analysis

↓

Evidence Verification

↓

LLM Explanation

This experimental structure is essential because otherwise the project would only demonstrate that an AI application can be built.

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20. Evaluation Metrics

The evaluation will measure several dimensions.

Applicability

- Precision
- Recall
- F1-score

Regulatory Change Detection

- Precision
- Recall
- F1-score

Obligation Identification

- Precision
- Recall
- F1-score

Impact Identification

- Precision

- Recall
- F1-score

Evidence Retrieval

- Evidence precision
- Evidence recall
- source correctness
- section/page correctness.

Generation Quality

- unsupported claim rate
- hallucination rate
- citation correctness.

Human Evaluation

If feasible, domain-informed evaluators may assess:

- usefulness
- correctness
- clarity
- confidence
- task completion time.

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21. Technology Stack

Programming Language — Python

Python will be the primary language for the AI and research pipeline.

Reasons:

- strong NLP ecosystem
- strong machine-learning ecosystem
- easy experimentation
- mature testing support
- integration with LLM and embedding APIs
- suitable for data processing.

Backend — FastAPI

FastAPI will expose the analysis functionality as APIs.

Example:

POST /businesses

POST /regulatory-analysis

GET /regulations

GET /changes

GET /evidence

GET /evaluation

FastAPI is appropriate because the project requires a clean separation between AI/research logic and frontend presentation.

Frontend — React + TypeScript

The frontend will provide:

- business-profile input
- regulatory-change selection
- analysis execution
- impact report
- evidence display
- experiment/evaluation dashboard.

TypeScript will improve maintainability and type safety.

Database — PostgreSQL

PostgreSQL will store structured information such as:

- businesses
- regulatory metadata
- versions
- obligations

- scenarios
- analysis results
- experiment results.

Vector Storage — pgvector or Qdrant

A vector database/index will store embeddings for semantic retrieval.

The final choice will depend on implementation and deployment constraints.

For an initial system, PostgreSQL + pgvector can reduce architectural complexity.

A dedicated vector database such as Qdrant can be evaluated if retrieval scale or functionality justifies it.

Embedding Model

An embedding model will convert regulatory passages into vector representations.

The selection will be based on:

- semantic retrieval quality
- computational cost
- latency
- language/domain suitability
- reproducibility.

The project should avoid choosing a model merely because it is popular.

LLM

The LLM will be used primarily for:

- evidence interpretation
- structured reasoning support
- impact explanation
- action generation
- natural-language reporting.

The LLM will not be treated as the authoritative source of regulatory facts.

The authoritative source will remain the regulatory corpus.

Reranker

A reranking model will be used to improve the ordering of retrieved regulatory passages.

This is particularly important when multiple regulations contain similar terminology.

Document Processing

Potential tools:

- PyMuPDF
- pdfplumber
- OCR where necessary
- Python text-processing libraries.

The actual tool will depend on the quality and structure of the collected documents.

Machine Learning / NLP

Potential technologies include:

- Hugging Face Transformers
- sentence-transformers
- scikit-learn
- NumPy
- pandas.

These will support embeddings, classification/evaluation, data processing, and experimentation.

Testing

pytest

Testing will cover:

- data validation
- document processing

- retrieval
- applicability logic
- API behaviour
- evaluation calculations.

DevOps

The final engineering system can use:

Git

GitHub

Docker

GitHub Actions

for:

- version control
- reproducible environments
- automated testing
- CI/CD
- deployment.

This also strengthens the project as an AI Engineering/Software Engineering portfolio project.

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22. Why This Technology Stack?

The architecture intentionally avoids unnecessary technology.

For example:

Knowledge graph is not mandatory.

Agentic workflows are not mandatory.

Multiple LLMs are not mandatory.

Microservices are not mandatory.

The system should first establish the research result.

The philosophy is:

Research Problem

↓

Research Method

↓

Required Technology

rather than:

Interesting Technology

↓

Find a problem for it

This keeps the project academically defensible.

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23. Feasibility Analysis

23.1 Technical Feasibility — HIGH

The system is technically feasible because its components can be implemented using mature technologies:

- Python
- PostgreSQL
- vector retrieval
- embedding models
- LLM APIs/open models
- FastAPI
- React
- Docker.

The project does not require training a foundation model from scratch.

The research contribution lies primarily in:

representation + retrieval + context-aware reasoning + evaluation

rather than foundation-model training.

Assessment

High feasibility

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24. Data Feasibility — MEDIUM/HIGH

Data availability is one of the most important risks.

However, the project has a manageable strategy because it does not require millions of documents.

The initial target is:

50–100 authoritative regulatory documents/versions

and:

100–300 structured benchmark scenarios.

Official Sri Lankan sources already provide relevant material. For example, the FCAU publishes current food regulations, including multiple labelling, food-safety and product-related regulations. The Ministry of Industry publishes industry-registration requirements and associated documentation.

The challenge is not merely finding documents.

The difficult part is creating **high-quality ground truth** for applicability and impact.

Therefore, dataset construction will be treated as a research activity rather than a simple scraping task.

Assessment

Medium to high feasibility

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25. Research Feasibility — HIGH

The project has a clear experimental structure:

Keyword

vs

Vector RAG

vs

Hybrid RAG

vs

Context-Aware Proposed System

This makes it possible to measure whether the proposed approach provides an improvement.

The project therefore has a research question that can be tested experimentally rather than merely demonstrated.

Assessment

High feasibility

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26. Computational Feasibility — HIGH

The project does not require large-scale model training.

Most computationally expensive tasks can be handled through:

- pretrained embedding models
- API-based LLMs
- lightweight rerankers
- local development
- small-to-medium document collections.

The benchmark size is also intentionally manageable.

Therefore, the project can be developed incrementally on ordinary development hardware, with cloud/API resources used where necessary.

Assessment

High feasibility

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27. Financial Feasibility — HIGH

The project can be developed using open-source components.

Potential costs include:

- LLM API usage
- cloud hosting
- optional database hosting
- deployment.

During development, costs can be controlled by:

- using small datasets initially
- caching embeddings
- using local/open models where practical

- limiting LLM calls
- running experiments in batches.

Assessment

High feasibility

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28. Time Feasibility — MEDIUM/HIGH

The project is feasible within an undergraduate project timeline if scope is controlled.

The biggest danger is uncontrolled expansion.

The project should NOT attempt to support:

- every Sri Lankan regulation
- every industry
- every government agency
- legal advice
- automated government submissions
- a nationwide compliance platform.

Instead:

Sri Lanka

↓

Food Manufacturing

↓

Small Businesses

↓

Selected Regulatory Domains

↓

Regulatory Change Impact

This keeps the research manageable.

Assessment

Medium to high feasibility

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29. Operational Feasibility — HIGH

The intended workflow is understandable to a business user:

Enter Business Profile

↓

Select/Receive Regulatory Change

↓

Analyze

↓

View Impact

↓

Inspect Evidence

The output is designed around business questions rather than technical AI terminology.

Assessment

High feasibility

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30. Legal and Ethical Feasibility

The system must explicitly be positioned as:

AI-assisted regulatory information and decision support

rather than legal advice.

The system should:

- use authoritative sources
- display source evidence
- display effective dates
- identify uncertainty
- avoid unsupported claims
- avoid pretending to make legal determinations
- preserve source versions
- distinguish historical and current regulations.

Human verification remains necessary for real-world compliance decisions.

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31. Overall Feasibility Matrix

Dimension	Assessment
Technical feasibility	**High**
Data feasibility	**Medium–High**
Research feasibility	**High**
Computational feasibility	**High**
Financial feasibility	**High**
Time feasibility	**Medium–High**
Operational feasibility	**High**
Ethical/legal feasibility	**High with safeguards**
Overall	**Feasible**

The largest risk is **not technology**.

The largest risk is:

Building an inaccurate or weakly validated regulatory benchmark.

Therefore, regulatory-source verification and benchmark quality must receive more attention than flashy AI features.

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32. Innovation

The project does not claim that the following concepts are individually novel:

- RAG
- hybrid retrieval
- regulatory change detection
- knowledge graphs
- LLM-based compliance systems
- applicability analysis.

Instead, the proposed contribution is the investigation of a specific combination:

Business-context-aware regulatory change impact analysis for Sri Lankan small food manufacturers, evaluated through controlled benchmarks against conventional retrieval approaches.

The contribution is therefore expected to be strongest in:

1. the Sri Lankan domain
2. the business-context representation
3. the regulatory-impact benchmark
4. contextual applicability reasoning
5. evidence-grounded impact analysis
6. experimental comparison against baseline RAG approaches.

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33. Expected Research Contribution

The project is expected to contribute:

Contribution 1 — Context Model

A structured representation of business characteristics relevant to regulatory applicability.

Contribution 2 — Regulatory Impact Framework

A framework connecting:

Regulatory Change

↓

Applicability

↓

Obligation

↓

Business Impact

↓

Action

↓

Evidence

Contribution 3 — Benchmark Dataset

A structured Sri Lankan regulatory impact benchmark for small food manufacturers.

Contribution 4 — Experimental Evidence

Quantitative comparison between:

- keyword retrieval
- vector RAG
- hybrid RAG
- context-aware regulatory reasoning.

Contribution 5 — Working AI Engineering Prototype

A practical system demonstrating how the research can be transformed into a usable software architecture.

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34. Expected Software Output

The final system should contain:

Frontend

- business profile management
- regulatory-change selection
- impact analysis
- evidence display
- analysis history
- evaluation dashboard.

Backend

- REST APIs
- business context service
- regulatory knowledge service
- retrieval service
- applicability engine
- impact engine
- evidence service
- LLM orchestration.

AI Pipeline

Document

↓

Processing

↓

Knowledge Representation

↓

Embedding

↓

Hybrid Retrieval

↓

Context Filtering

↓

Applicability Reasoning

↓

Impact Analysis

↓

Evidence Validation

↓

LLM Explanation

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35. Expected Final Output

The final report generated for a business should follow a consistent structure:

REGULATORY CHANGE

What changed?

APPLICABILITY

Applicable / Not Applicable /

Conditional

REASON

Which business characteristics

caused this result?

AFFECTED OBLIGATION

What requirement is relevant?

BUSINESS IMPACT

What operation/product/document

is affected?

REQUIRED ACTION

What should the business review

or change?

EFFECTIVE DATE

When does the change apply?

EVIDENCE

Official regulation

Section / Clause

Page

Source

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36. Project Limitations

The project will have several limitations.

1. Limited Regulatory Coverage

The initial system will not cover every Sri Lankan regulation.

2. Limited Industry Coverage

The research focuses on small food manufacturers.

3. Regulatory Interpretation

The system will not replace legal professionals or regulatory authorities.

4. Document Quality

Some historical documents may be difficult to extract accurately.

5. Language

The initial research implementation may prioritize English-language regulatory documents, with multilingual support considered as future work.

6. Ground Truth

Some regulatory-impact interpretations may require expert validation.

7. LLM Dependence

LLM output can contain errors; therefore evidence verification and structured evaluation are necessary.

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37. Risk Management

Risk	Impact	Mitigation
Insufficient regulatory documents	High	Narrow domain and prioritize authoritative sources
Incorrect ground truth	Very High	Source verification and expert review
PDF extraction errors	Medium	Multiple extraction methods + manual verification

Risk	Impact	Mitigation
LLM hallucination	High	Evidence-grounded generation + validation
Poor retrieval	High	Hybrid retrieval + reranking
Scope expansion	Very High	Strict food-manufacturing scope
API cost	Medium	Caching + limited experiments + local models
Time constraints	High	Phase-based development
Regulation changes during project	Medium	Versioned corpus and effective dates
Overengineering	High	Research-first architecture

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38. Success Criteria

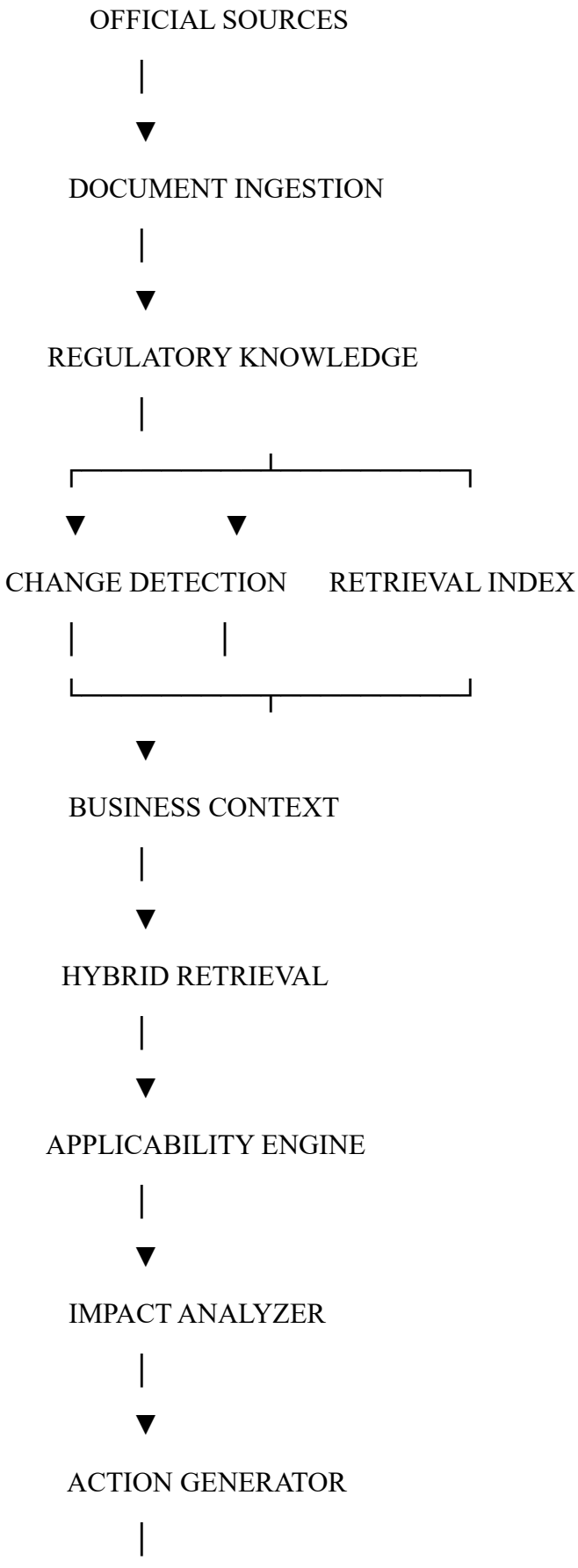
The project will be considered successful if:

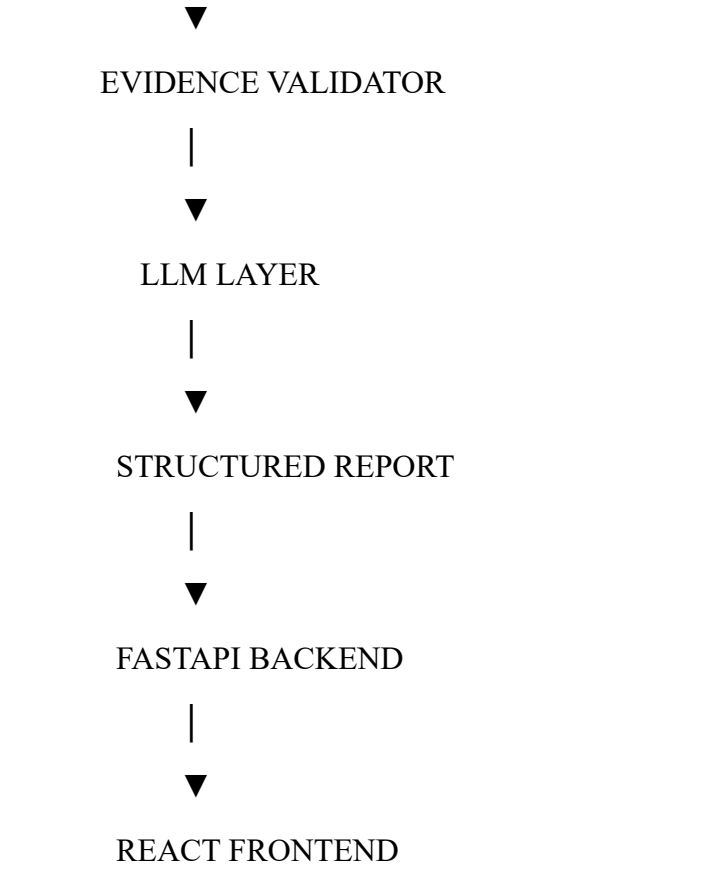
1. A reliable regulatory corpus is created.
2. Regulatory versions and effective dates can be represented.
3. Business context can be represented structurally.
4. Regulatory changes can be identified.
5. Keyword, vector, and hybrid baselines can be implemented.
6. The proposed context-aware system can be evaluated.
7. Applicability can be measured quantitatively.
8. Impact identification can be measured quantitatively.
9. Evidence correctness can be evaluated.
10. Hallucination/unsupported-claim rates can be measured.
11. The proposed approach demonstrates a meaningful improvement or produces useful findings even where improvement is limited.
12. The complete system can be reproduced by another developer.
13. The final research question can be answered using experimental evidence.

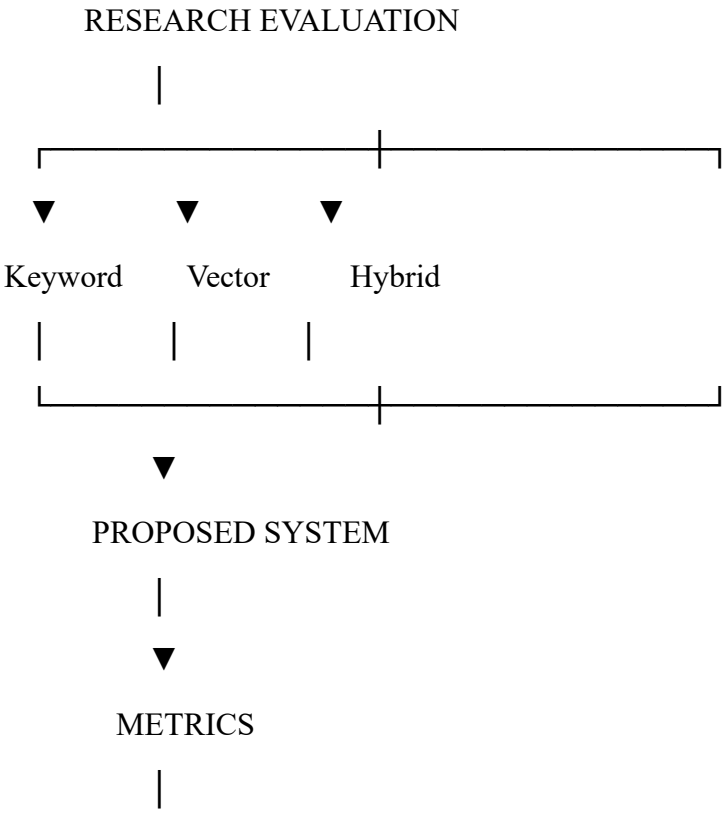
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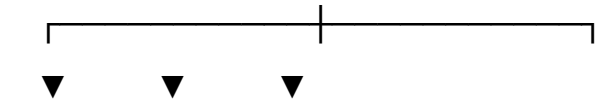
39. Expected Final Architecture

The completed project will follow:




EVIDENCE VALIDATOR
LLM LAYER
STRUCTURED REPORT
FASTAPI BACKEND
REACT FRONTEND


RESEARCH EVALUATION
Keyword
Vector
Hybrid
PROPOSED SYSTEM
METRICS



Applicability Evidence Hallucination

F1 Accuracy Rate

---

40. Future Extensions

If the core research succeeds, the system could later be extended to:

- multilingual Sinhala/Tamil regulatory analysis
- additional SME industries
- automated regulatory-change monitoring
- knowledge graphs
- regulation-to-regulation dependency analysis
- compliance calendars
- notification systems
- regulatory risk scoring
- document comparison
- enterprise compliance management
- human-in-the-loop expert review
- continuous regulatory monitoring.

These features are deliberately outside the initial scope.

---

41. Final Expected Outcome

The project is expected to produce two complementary outcomes.

Research Outcome

A validated investigation into whether **business-context-aware regulatory retrieval and reasoning** improves regulatory applicability and impact analysis compared with conventional RAG approaches.

Engineering Outcome

A working AI engineering prototype that demonstrates:

- document processing

- structured knowledge representation
- hybrid information retrieval
- embeddings
- reranking
- LLM integration
- evidence verification
- REST APIs
- database integration
- frontend development
- automated testing
- Dockerization
- CI/CD
- experiment evaluation.

---

42. Conclusion

RegImpact-SL addresses a real information problem faced by businesses operating within a fragmented and changing regulatory environment.

Rather than building a generic legal chatbot, the project focuses on a narrower and experimentally testable problem:

Given a regulatory change and the characteristics of a specific small food manufacturer, can an AI system accurately determine whether the change applies, explain why, identify the resulting business impact and required action, and support those conclusions with authoritative evidence?

The project is technically feasible because it can be constructed using established AI, NLP, retrieval, database, and software-engineering technologies without training a foundation model from scratch.

It is research-feasible because the proposed system can be compared against controlled baselines using a dedicated benchmark.

It is practically relevant because Sri Lankan SMEs operate within a complex regulatory and institutional environment, while current official sources demonstrate that food and manufacturing businesses must deal with multiple regulatory instruments and requirements.

Most importantly, the project maintains a clear separation between the **research problem** and the **technology used to solve it**.

The research problem is:

Context-aware regulatory change impact analysis.

The technology is:

Structured regulatory knowledge + hybrid retrieval + LLM reasoning + evidence verification.

Therefore, the project can be evaluated not by asking whether "AI works," but by asking a much stronger research question:

Does incorporating business context improve the accuracy and evidence-groundedness of regulatory impact analysis compared with conventional RAG?

That is the central proposition of RegImpact-SL.

Evidence base used for the proposal

The proposal's Sri Lankan regulatory context is grounded in current official sources. The Ministry of Industry's current industry-registration information states that manufacturing production sites generally need registration within three months and lists supporting documentation. (Industry & Entrepreneurship Ministry)

The Ministry's **National SME Strategy Framework 2026** provides the broader policy context for SME development and regulatory/institutional improvement in Sri Lanka. (Industry & Entrepreneurship Ministry) ADB sources likewise describe the substantial economic role of Sri Lankan SMEs. (Asian Development Bank)

For the food-manufacturing domain, the official FCAU site currently lists multiple food regulations, including the **Food (Labelling and Advertising) Regulations 2026**, while the 2026 regulation states that it comes into operation on **1 July 2026**. (Directorate of Environmental Health) FCAU also documents a domestic food-control process involving application, inspection, sampling, testing, and certification. (Directorate of Environmental Health)

The environmental component is also grounded in the CEA's current Environmental Protection Licence framework, including its prescribed-activity and local-authority arrangements. (Central Environmental Authority)

My recommendation: use this as the **master proposal**, but don't submit it unchanged yet. The next important step is to turn it into your university's required proposal format—typically **abstract → background → problem statement → gap → RQs → objectives → methodology → feasibility → timeline → references**—and, importantly, we should make the literature-gap section stronger with a focused review of existing regulatory RAG/compliance research.

ARCHITECTURE PLAN

Absolutely. For **RegImpact-SL**, we should design the architecture around the **research problem first**, and then place AI/RAG/software-engineering components around it.

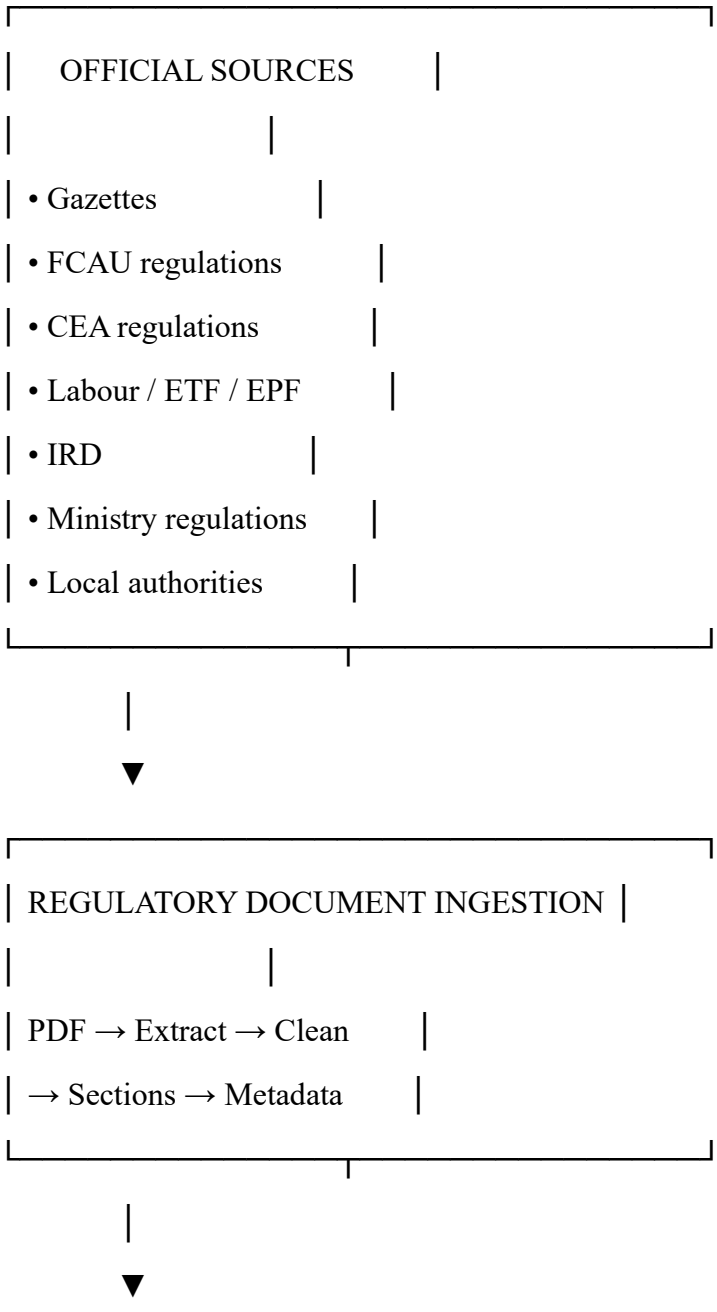
The key principle is:

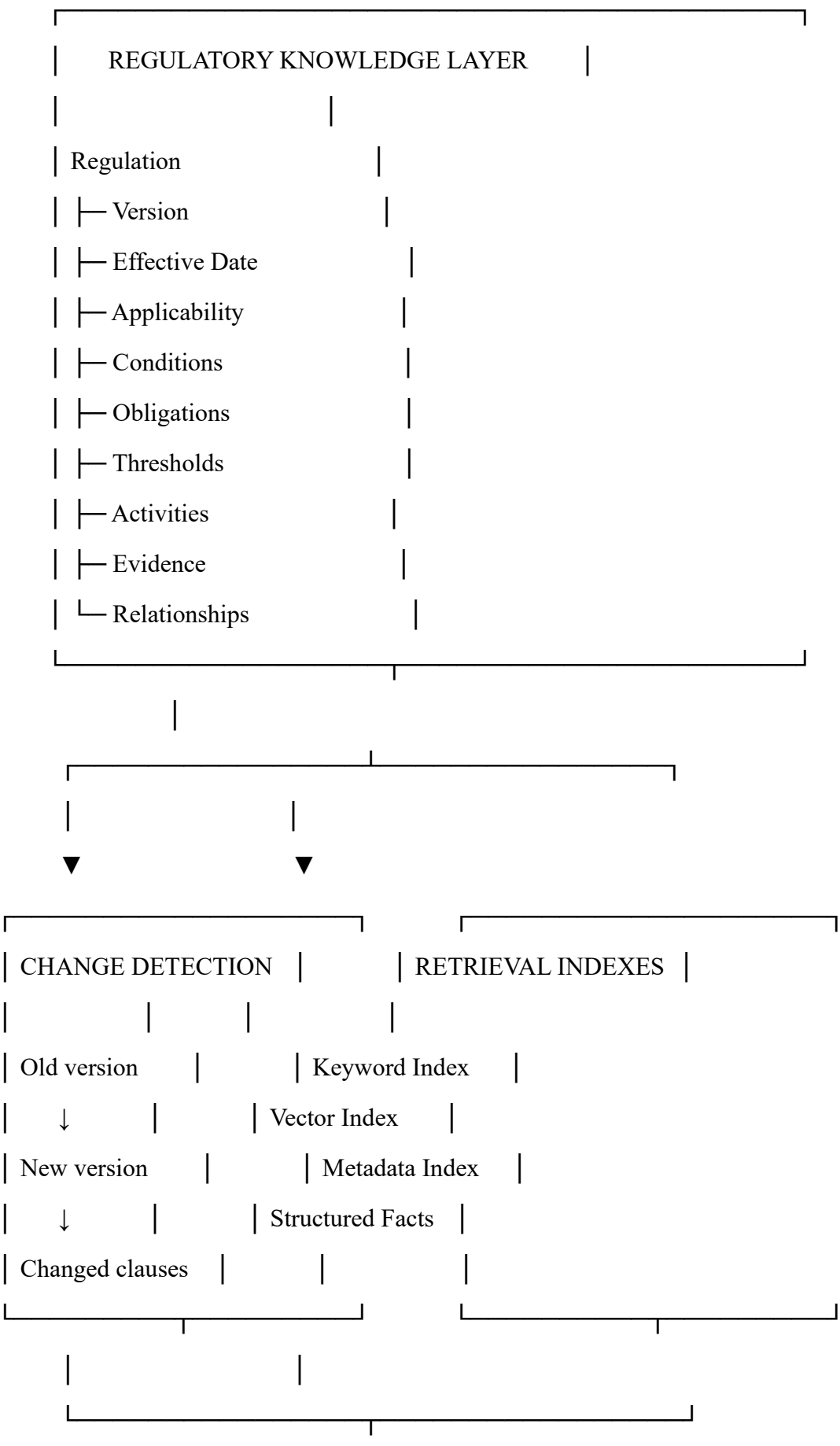
Business Context + Regulatory Change → Applicability → Impact → Action → Evidence

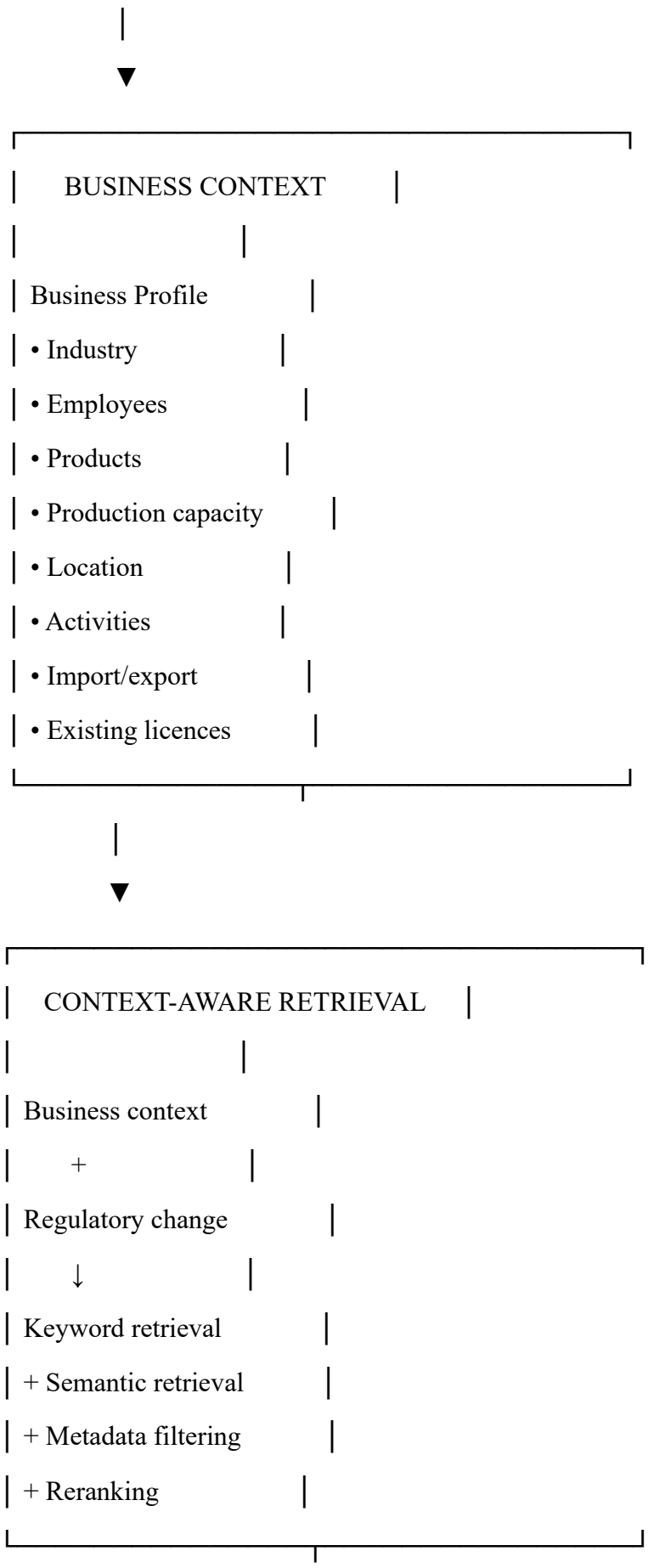
Not simply:

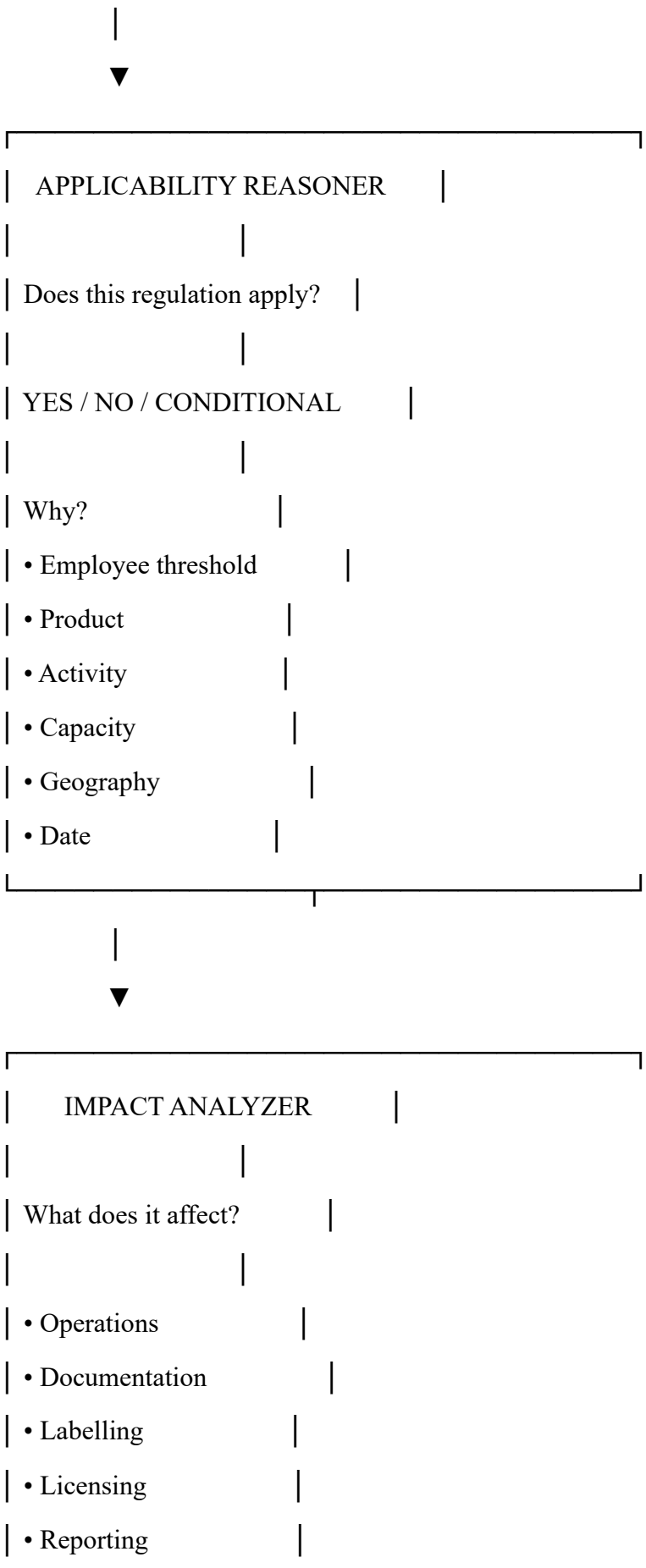
User → Chatbot → LLM → Answer

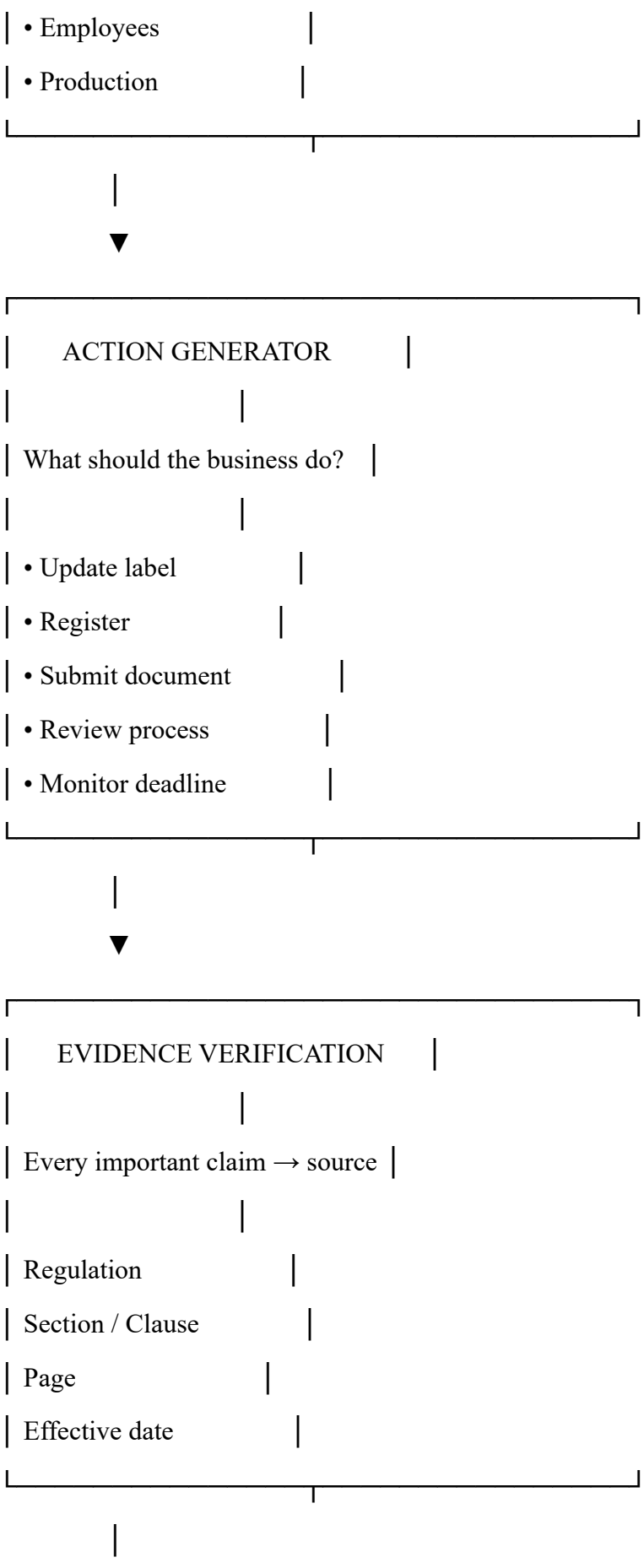
1. Entire System — High-Level Architecture

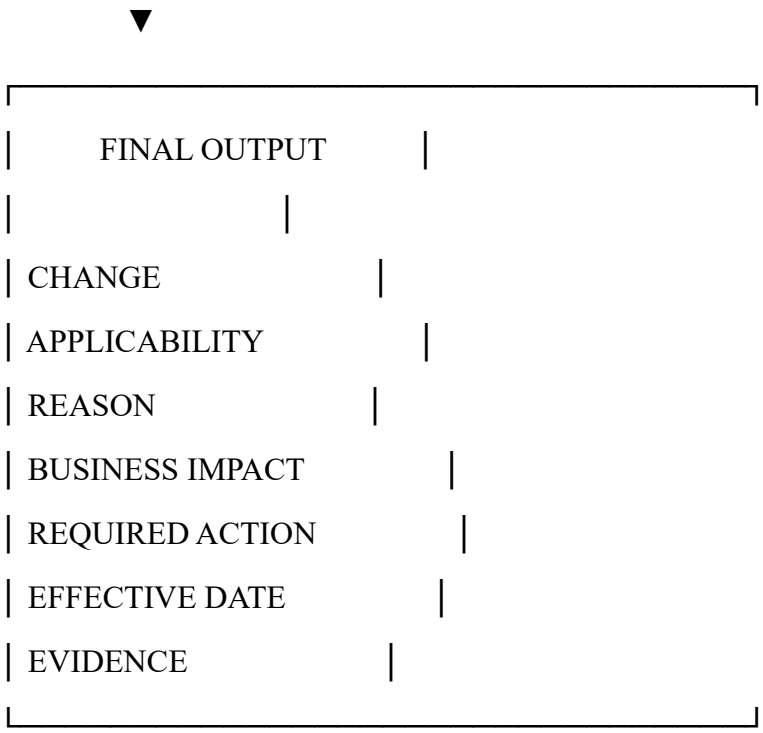



REGULATORY KNOWLEDGE LAYER
Regulation
Version
Effective Date
Applicability
Conditions
Obligations
Thresholds
Activities
Evidence
Relationships
CHANGE DETECTION
Old version
New version
Changed clauses
RETRIEVAL INDEXES
Keyword Index
Vector Index
Metadata Index
Structured Facts


BUSINESS CONTEXT
Business Profile
• Industry
• Employees
• Products
• Production capacity
• Location
• Activities
• Import/export
• Existing licences
CONTEXT-AWARE RETRIEVAL
Business context
+
Regulatory change
↓
Keyword retrieval
+ Semantic retrieval
+ Metadata filtering
+ Reranking


APPLICABILITY REASONER
Does this regulation apply?
YES / NO / CONDITIONAL
Why?
• Employee threshold
• Product
• Activity
• Capacity
• Geography
• Date
IMPACT ANALYZER
What does it affect?
• Operations
• Documentation
• Labelling
• Licensing
• Reporting


• Employees
• Production
ACTION GENERATOR
What should the business do?
• Update label
• Register
• Submit document
• Review process
• Monitor deadline
EVIDENCE VERIFICATION
Every important claim → source
Regulation
Section / Clause
Page
Effective date



---

2. Architecture by Layers

I recommend thinking about the project as **8 major layers**.

Layer 1 — Source Layer

This is where the authoritative regulatory information originates.

Official Government Sources

- Gazettes
- Ministries
- FCAU
- CEA
- Department of Labour
- ETF
- IRD
- Local Authorities

Important: the system should not treat random websites or LLM knowledge as the authoritative source.

---

3. Regulatory Document Processing Layer

Official PDF



Document ingestion



Text extraction



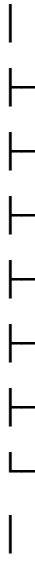
Cleaning



Section / clause detection



Metadata extraction



Processed Regulatory Document

This layer is extremely important for your research because later you need to answer:

"Where exactly did the system get this answer?"

Not merely:

"The LLM says so."

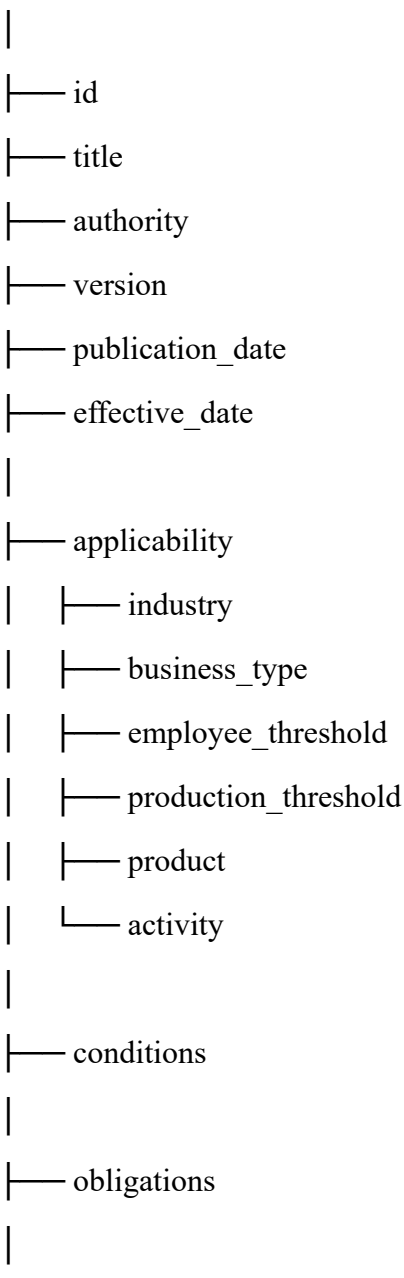
---

4. Regulatory Knowledge Layer

This is where we turn raw regulatory text into structured knowledge.

For example:

Regulation



├── affected_operations

│

└── evidence

    ├── document

    ├── section

    ├── page

    └── source

Example

Suppose:

Business:

6 employees

Food manufacturer

Packaged confectionery

Domestic sales

And the regulation says:

Applies to food manufacturing establishments

employing 5 or more workers.

The structured representation becomes:

Regulation:

    target_activity = food manufacturing

    employee_min = 5

Business:

    activity = food manufacturing

    employees = 6

Condition:

    6 >= 5

Result:

APPLICABLE

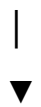
This structured layer is one of the things that differentiates the project from a basic RAG chatbot.

---

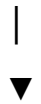
5. Regulatory Change Detection Layer

This is a separate component.

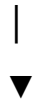
Old Regulation



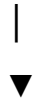
Version comparison



New Regulation



Clause-level comparison



Changed provisions

Output:


```
{
  "change_type": "threshold_change",
  "old_value": 10,
  "new_value": 15,
  "affected_condition": "employee_count",
  "effective_date": "...",
  "evidence": "..."
}
```

This becomes particularly powerful with your **minimal-pair experiments**.

For example:

Business A

14 employees

↓

NOT affected

Business B

15 employees

↓

AFFECTED

Same regulation.

Only one business attribute changes.

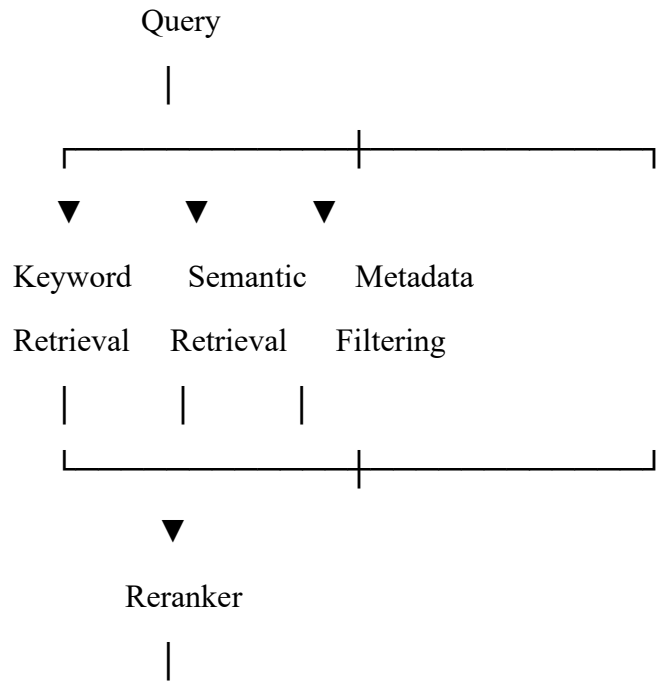
That is a very strong test of whether your system actually understands context.

---

6. Retrieval Layer

This should **not** be only vector search.

You need three retrieval mechanisms.



▼

Relevant evidence

Why?

Regulations contain things like:

- exact numbers
- dates
- Gazette numbers
- legal terminology
- section numbers
- thresholds
- names of agencies.

Pure semantic retrieval can miss exact regulatory details.

Keyword retrieval alone can miss semantic relationships.

Therefore:

Hybrid Retrieval = lexical + semantic + metadata

is much more defensible.

---

7. Context-Aware Reasoning Layer

This is the **heart of the project**.

The system receives:

BUSINESS CONTEXT

Industry = Food manufacturing

Employees = 6

Product = Biscuits

Production = 200 kg/day

Sales = Domestic

Import = No

Export = No

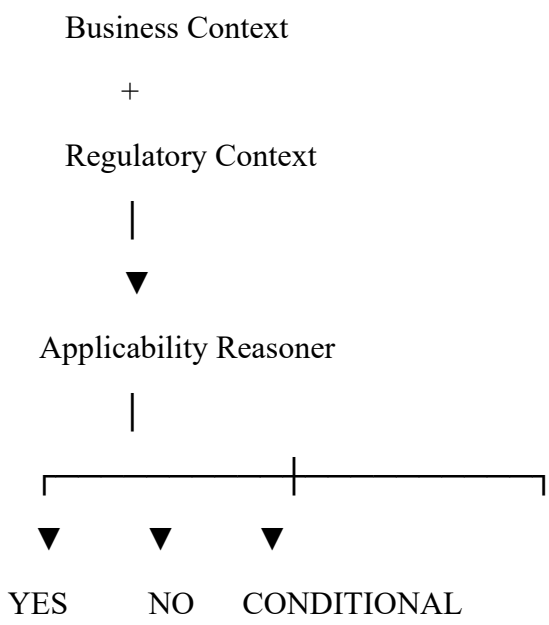
Location = Sri Lanka

and:

REGULATORY CHANGE

New regulation / changed provision

Then:



The reasoning engine should explicitly consider:

Employee conditions

Product conditions

Activity conditions

Capacity conditions

Geographic conditions

Business type

Effective date

Other regulatory conditions

---

8. Impact Analysis Layer

Once applicability is established:

Applicable Regulation



Affected Obligation



Business Operation



Business Impact

For example:

Regulatory change



New labelling requirement



Product packaging affected



Existing labels need review



Operational impact:

Packaging / QA / compliance

The system should distinguish:

Regulatory impact

"What obligation changed?"

from:

Business impact

"What does this mean for this particular company?"

That distinction is important.

---

9. Action Generation Layer

Then:

Business Impact

↓

Required Action

Example:

ACTION:

Review current packaging labels

against the new labelling requirements.

Priority:

High

Effective:

1 July 2026

Evidence:

Food (Labelling and Advertising)

Regulations 2026, Section X, Page Y

The LLM can help generate understandable explanations, but **the action must be grounded in retrieved regulatory evidence**.

---

10. Evidence Layer

This should be a first-class component rather than an afterthought.

Every significant output should have:

Claim

↓

Evidence

↓

Document

↓

Section

↓

Page

↓

Source

For example:

CLAIM

The requirement applies to this business.

EVIDENCE

Food Regulation 2026

SECTION

Section X

PAGE

Page 14

SOURCE

Official FCAU document

This gives you:

Evidence-grounded regulatory impact analysis

rather than merely AI-generated compliance text.

---

11. LLM Layer

The LLM should **not own the entire system**.

Instead:

Structured Facts

+

Retrieved Evidence

+

Business Context



LLM

Structured Explanation

The LLM's job:

- interpret retrieved evidence
- explain applicability
- summarize impact
- formulate actions
- communicate uncertainty
- produce human-readable output.

It should **not invent regulations**.

---

12. Knowledge Graph — Optional Research Extension

Do **not** make this mandatory in Phase 1.

Later architecture can include:

Business

├── operates_in → Industry

├── produces → Product

├── employs → EmployeeCount

│

└── performs → Activity

Regulation

│

├── applies_to → Industry

├── requires → Obligation

├── affects → Activity

├── replaces → PreviousVersion

└── effective_from → Date

Then:

Business

↓

Business Activity

↓

Applicable Regulation

↓

Obligation

↓

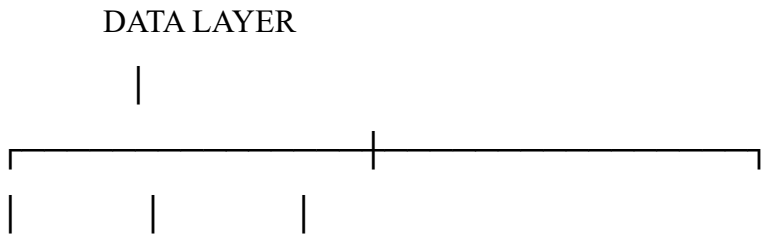
Business Impact

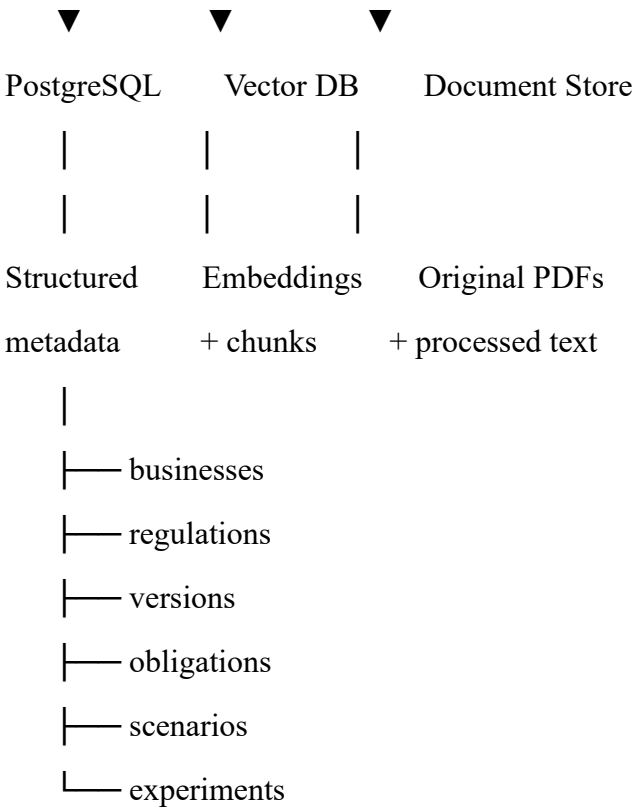
But this should be an **experimental extension**, not the foundation of the entire project.

---

13. Data Storage Architecture

I'd separate the data logically:





Possible implementation:

PostgreSQL

+

Qdrant / pgvector

+

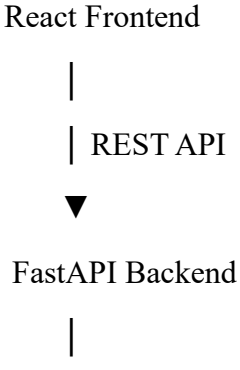
File/object storage

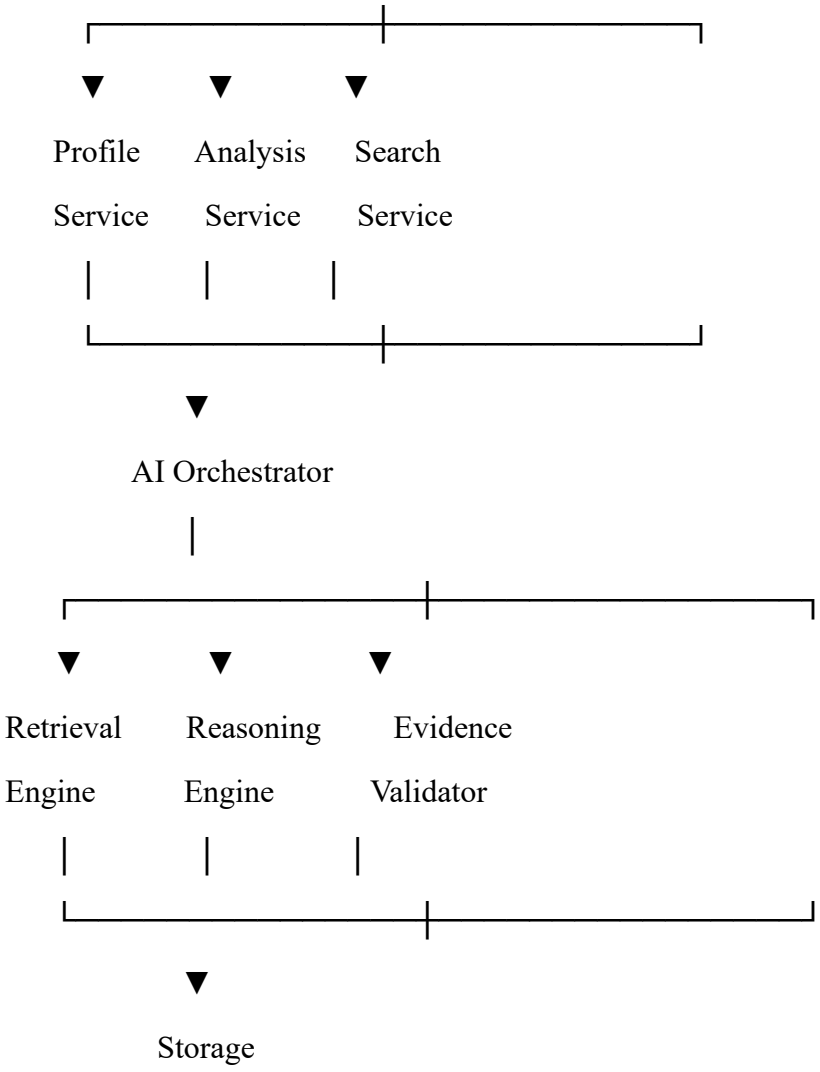
You don't need to decide the exact database today.

---

14. Backend Architecture

For the engineering implementation:





---

15. Frontend Architecture

The UI should make the research concept visible.

Screen 1 — Business Profile

Business Information

Industry: Food Manufacturing

Employees: 6

Products: Biscuits

Production: 200 kg/day

Import: No

Export: No

[Analyze Regulatory Impact]

Screen 2 — Regulatory Change

Regulation:

Food (Labelling and Advertising)

Regulations 2026

Effective Date:

1 July 2026

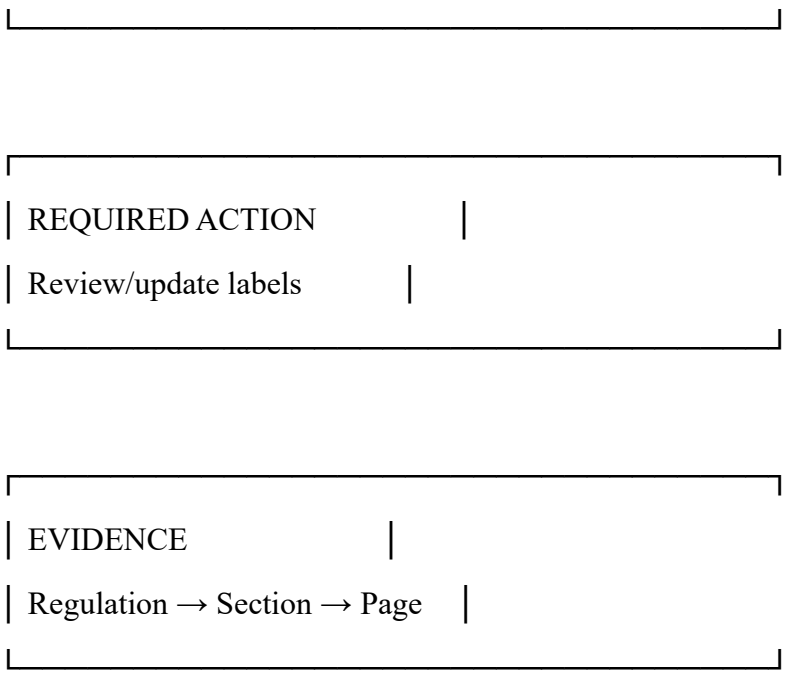
Screen 3 — Impact Report

┌──┐
│ REGULATORY CHANGE │
│ New labelling requirements │
└──┘

┌──┐
│ APPLICABILITY │
│ ✓ Applicable │
└──┘

┌──┐
│ WHY? │
│ Business produces packaged │
│ food products. │
└──┘

┌──┐
│ BUSINESS IMPACT │
│ Product packaging and labels │



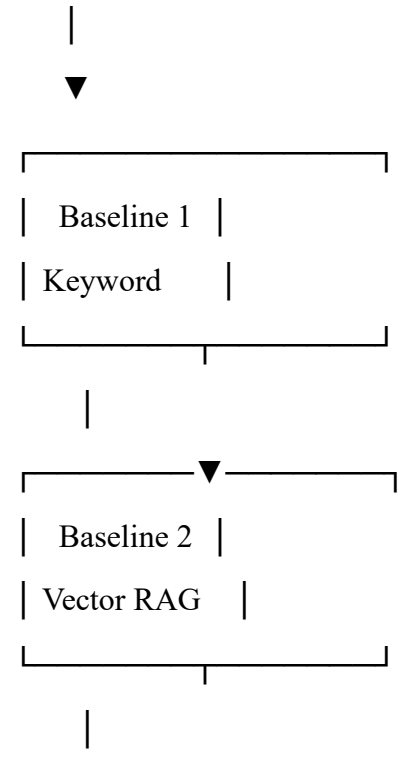
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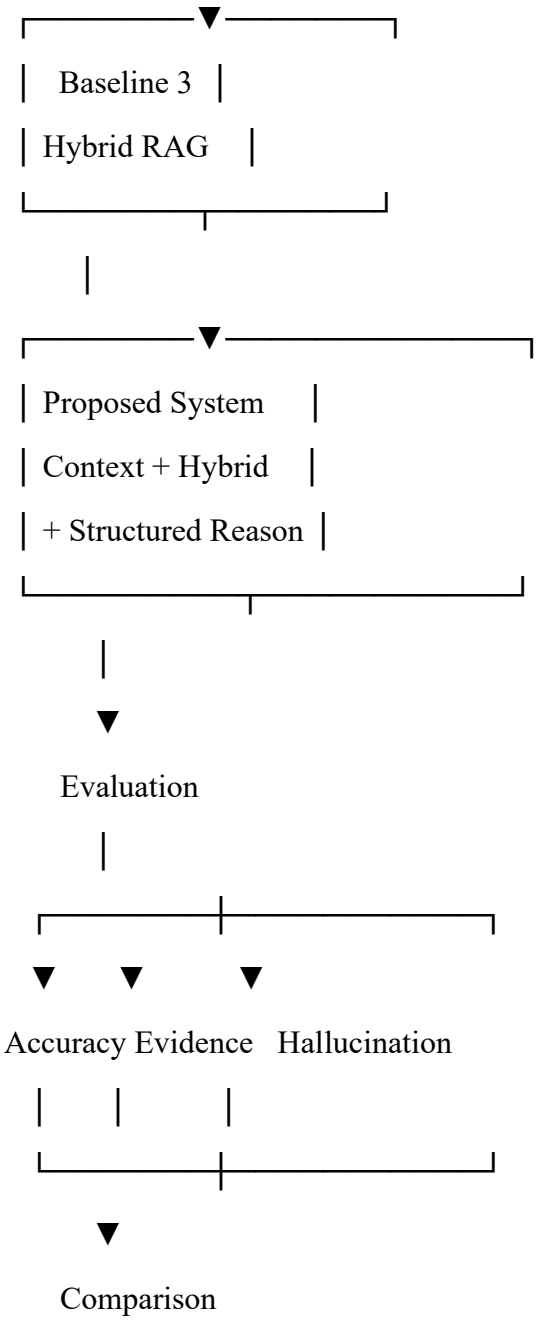
16. Research Experiment Architecture

This is extremely important.

The final application is **not the research experiment by itself**.

You need a separate evaluation pipeline:





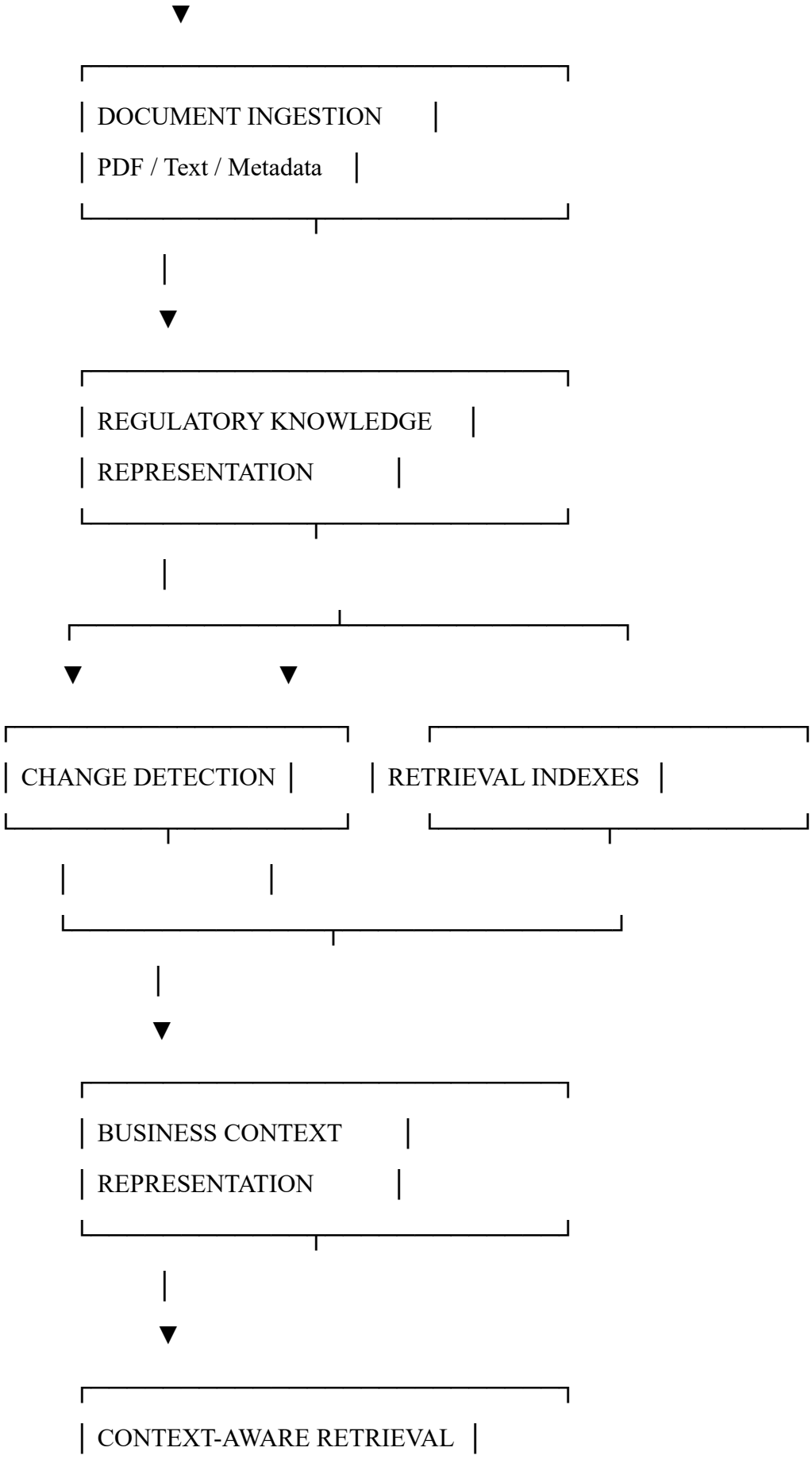
This is where the **research contribution** lives.

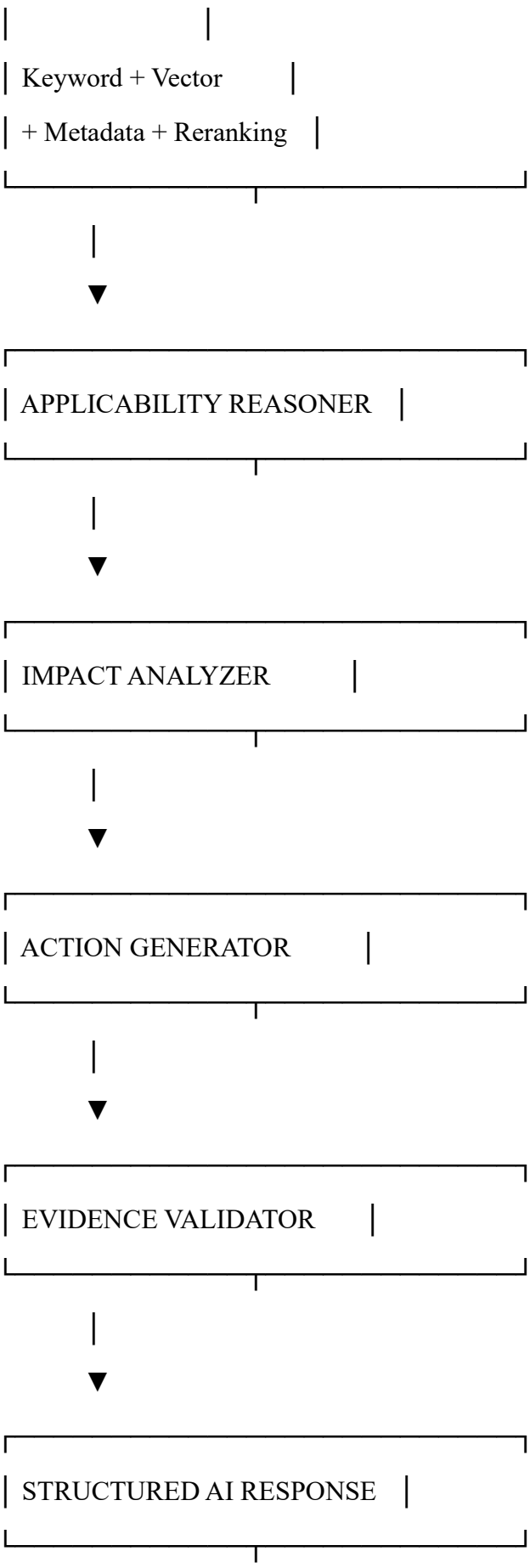
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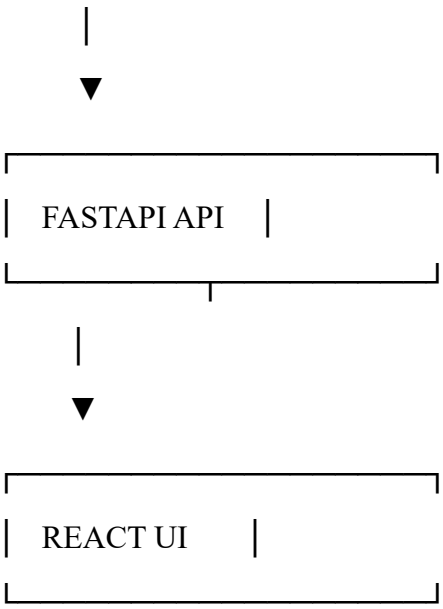
17. Complete Research/Engineering Architecture

Putting everything together:

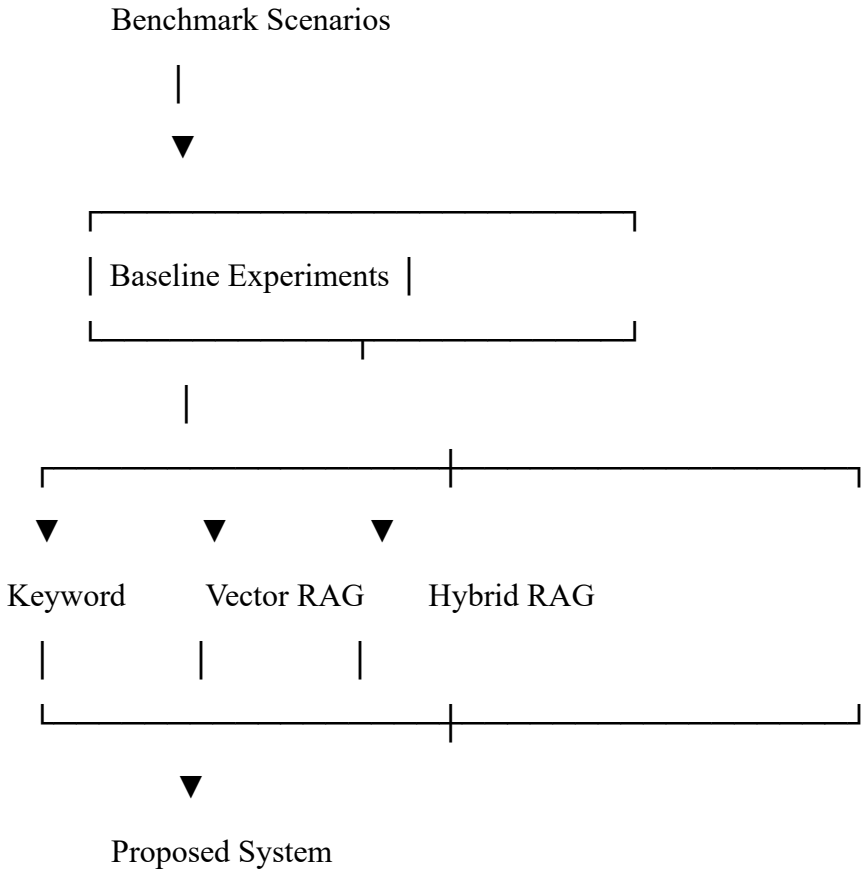


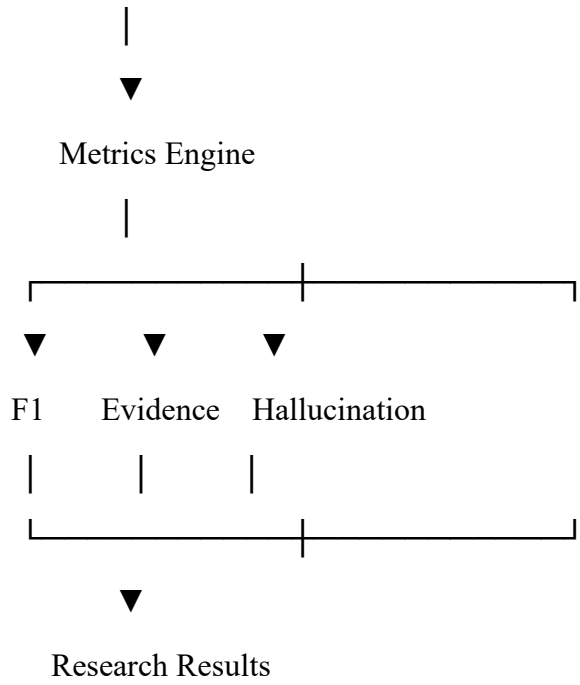

DOCUMENT INGESTION
PDF / Text / Metadata
REGULATORY KNOWLEDGE
REPRESENTATION
CHANGE DETECTION
RETRIEVAL INDEXES
BUSINESS CONTEXT
REPRESENTATION
CONTEXT-AWARE RETRIEVAL


Keyword + Vector
+ Metadata + Reranking
APPLICABILITY REASONER
IMPACT ANALYZER
ACTION GENERATOR
EVIDENCE VALIDATOR
STRUCTURED AI RESPONSE



PARALLEL RESEARCH EVALUATION PIPELINE





---

18. The Most Important Architectural Boundary

I want you to keep this distinction throughout the project:

Research architecture

Business Context

+

Regulatory Change

↓

Applicability

↓

Impact

↓

Action

↓

Evidence

Technology architecture

Python

FastAPI

PostgreSQL

Vector DB

Embeddings

Reranker

LLM

React

Docker

CI/CD

Do not reverse these.

The research question should determine the technology—not the other way around.

---

19. How This Maps to Our Implementation Phases

Phase	Architecture Component
1	Project foundation + data layer
2	Regulatory knowledge representation
3	Document processing
4	Keyword retrieval baseline
5	Vector RAG baseline
6	Hybrid RAG
7	Context-aware proposed system
8	Experimental evaluation
9	Production-quality AI engineering system
10	Research paper + portfolio

So **Phase 1 does not need to know how the final LLM system works internally**. We are deliberately building the foundation underneath it.

---

20. Final Architecture Goal

At the end, I want the project to be explainable in one sentence:

RegImpact-SL takes a specific small food manufacturer's business context and a regulatory change, retrieves the relevant authoritative provisions, determines whether the change applies, identifies its business impact and required action, and provides evidence-grounded explanations.

And from an engineering/interview perspective:

"I didn't build a chatbot. I built and evaluated a context-aware regulatory impact analysis pipeline with document ingestion, structured knowledge representation, hybrid retrieval, LLM reasoning, evidence verification, benchmark evaluation, API architecture, and a deployable frontend."

That is the architecture we should build toward. Phase 1 remains deliberately narrow; we should not implement this entire architecture at once.